

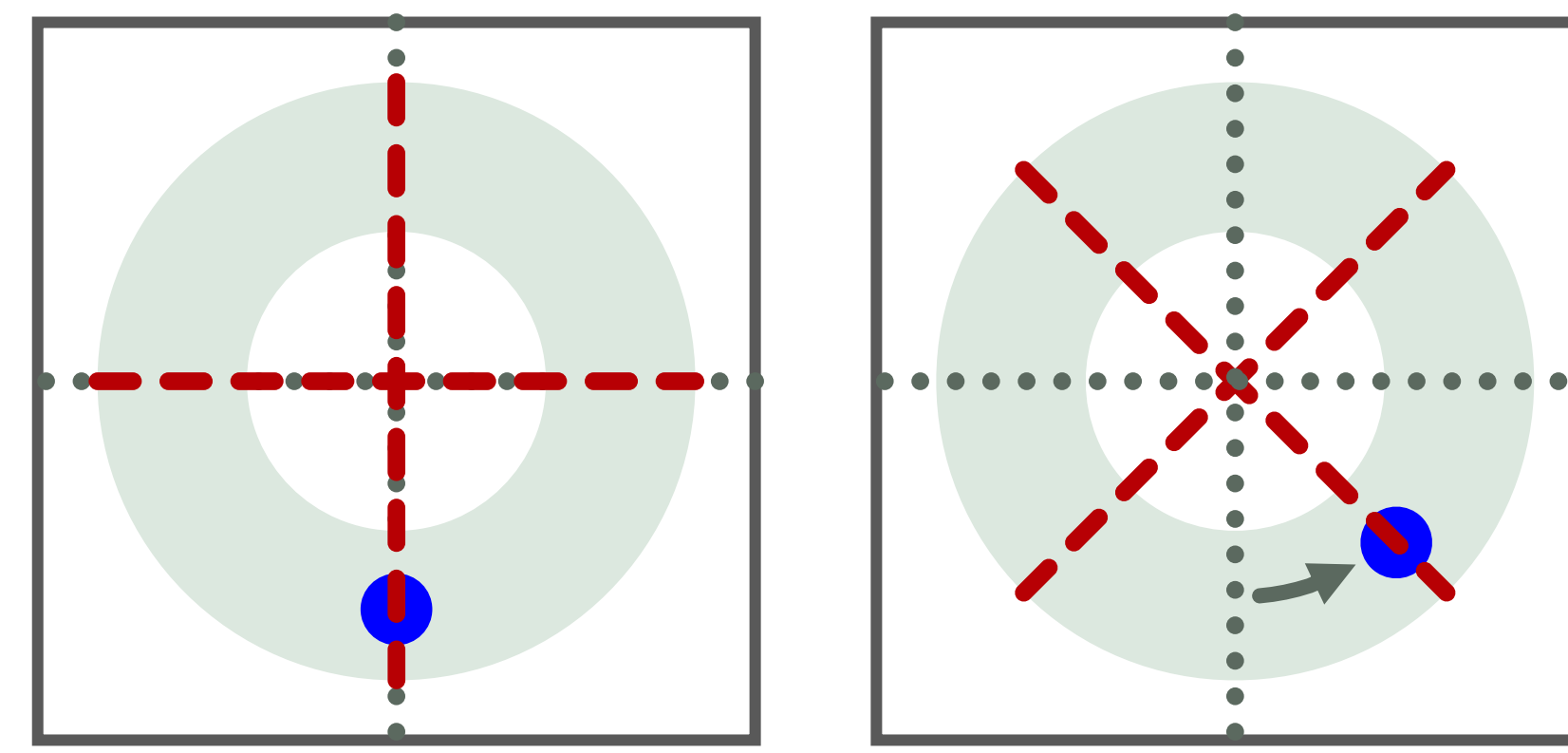
# Task-Irrelevant Motion Strengthens Motion Silencing of Color Changes

Hyerim Kim, Oakyoon Cha

Sogang University

## Introduction

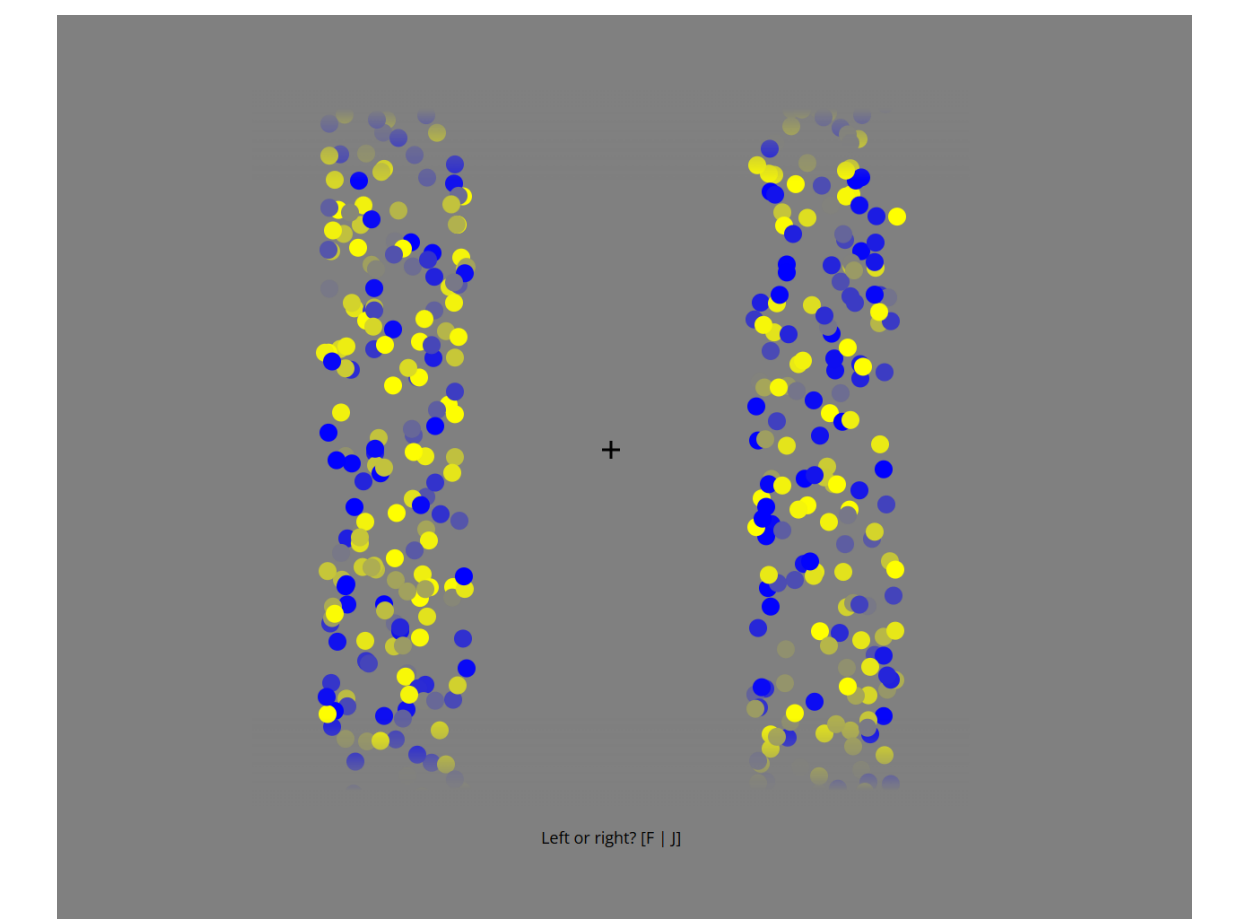
- **Motion Silencing Illusion** (Suchow & Alvarez, 2011)
- Wu and Flombaum (2024) suggested that motion silencing arises from conflict between multiple spatial coordinates caused by rotational motion.



## General method

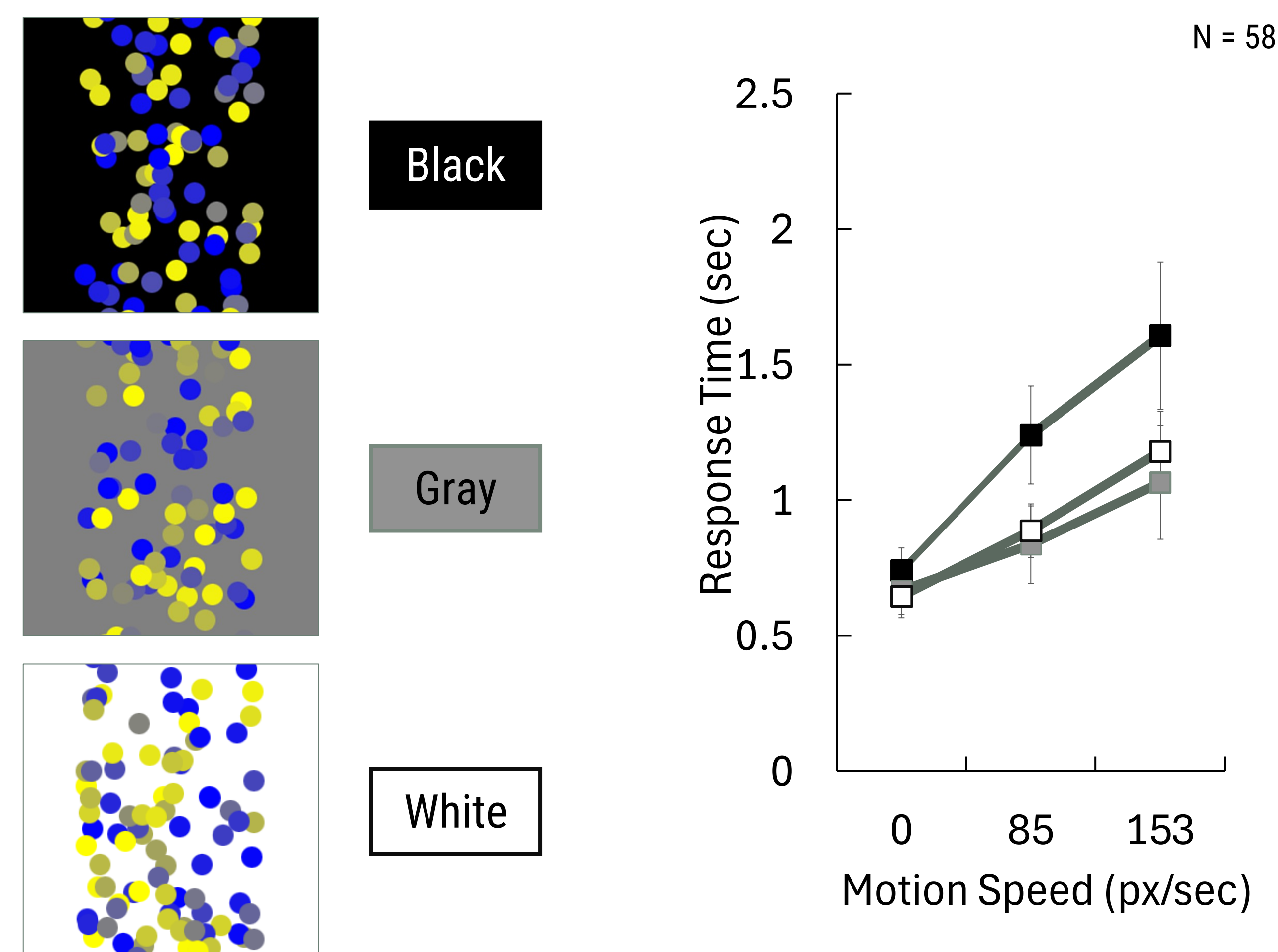
- The colors of dots in one array cycled between yellow and blue continuously, while those in the other array remained unchanged.
- Stronger motion silencing illusion would make it harder to distinguish which array is changing in color, resulting in longer response time (Wu & Flombaum, 2024).

Choose the array changing in color



## Experiment 1a

- Does background luminance influence the strength of the motion silencing illusion?

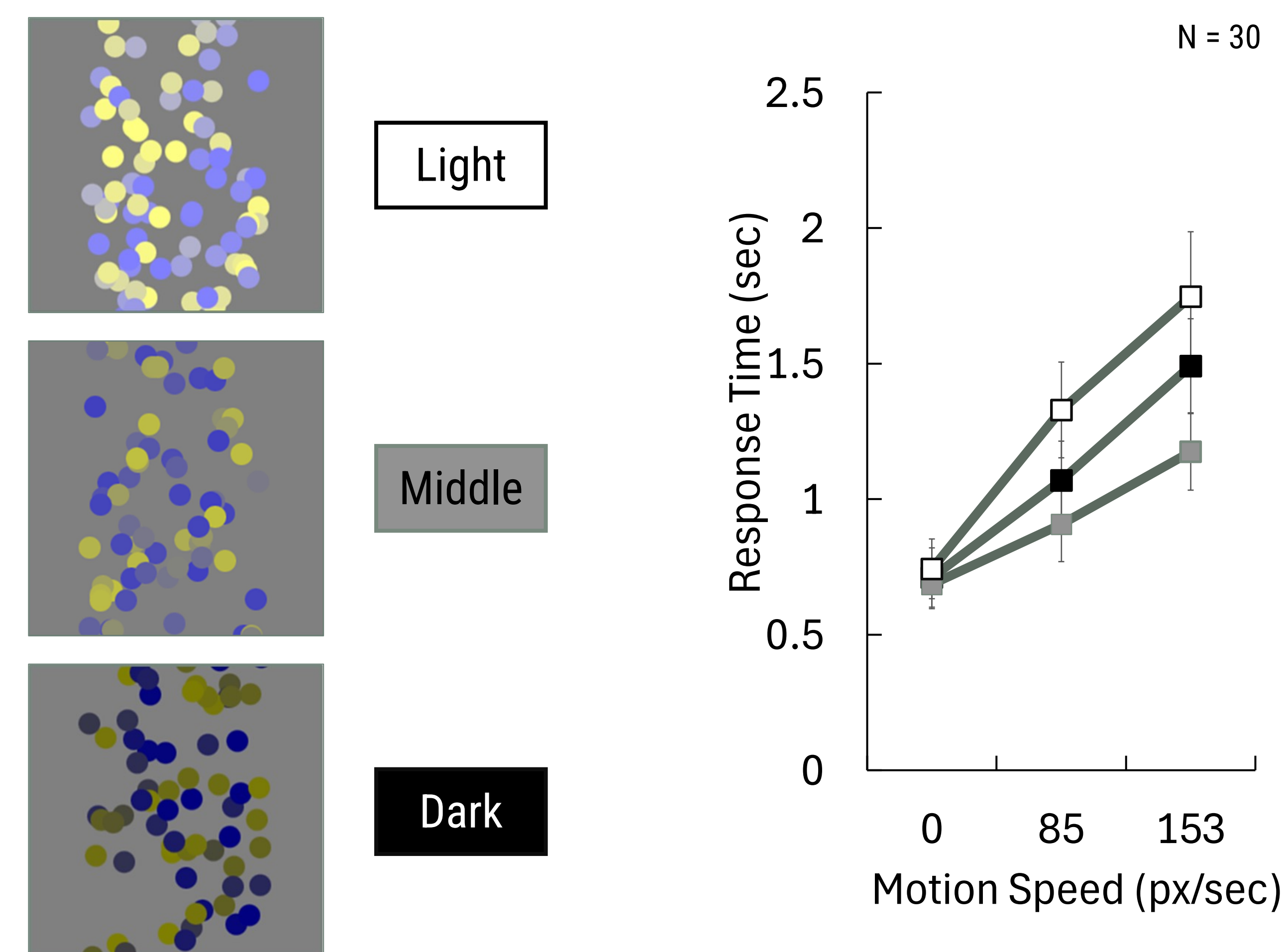


The weakest on **Gray** background  
The strongest on **Black** background

The motion silencing illusion was the weakest when the **contrast polarity reversal** occurred during color changes.

## Experiment 1b

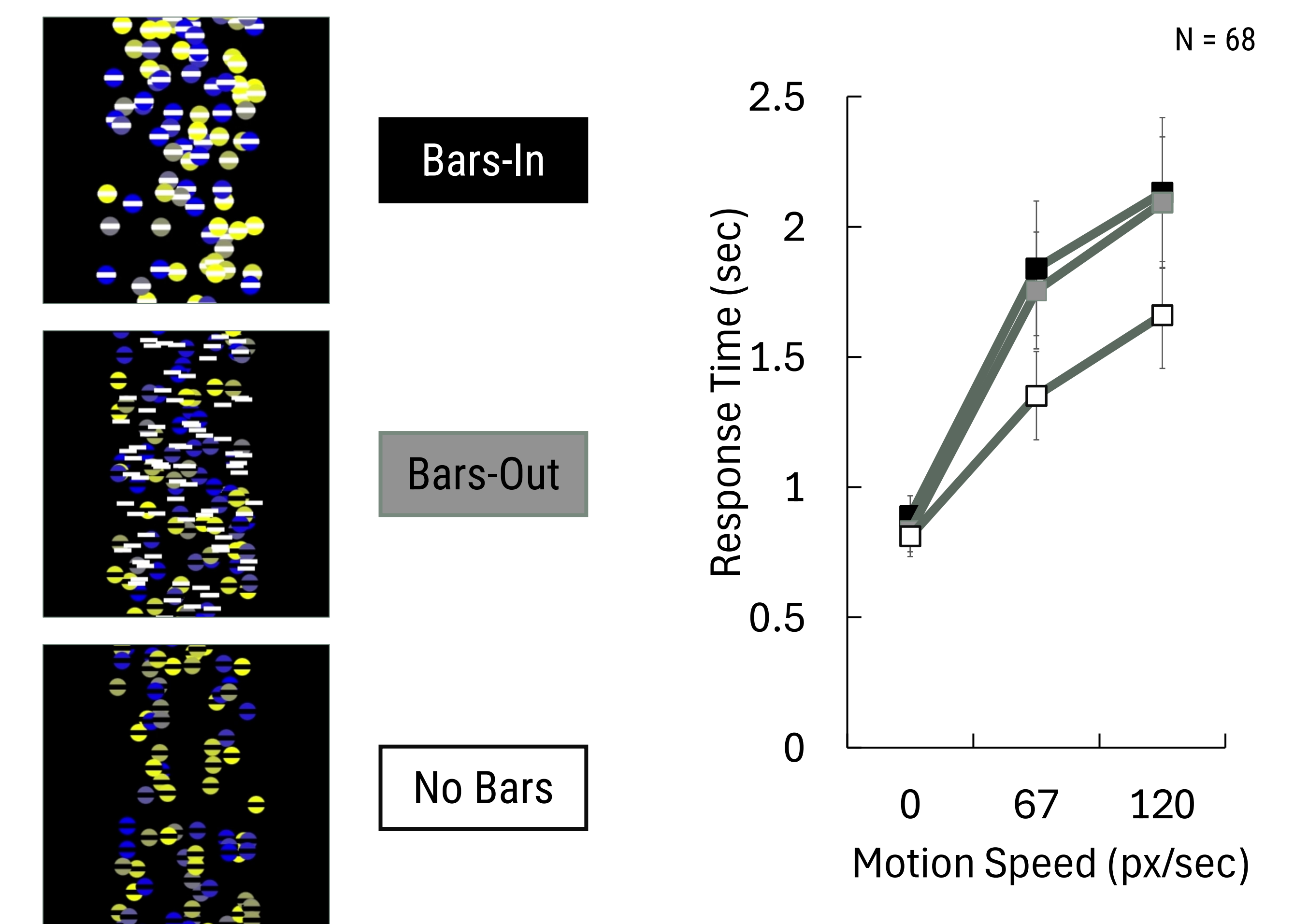
- Does the effect found in Experiment 1a arise from the luminance of the background itself or from the luminance contrast?



The weakest on the **Middle** condition  
The strongest on the **Light** condition

## Experiment 2

- Can task-irrelevant motion strengthen motion silencing illusion?



**Irrelevant motion signals** enhance the motion silencing effect, even when they originate from stimuli other than those undergoing color changes.

## Discussion

- The motion silencing effect is sensitive to the strength of low-level motion signals.
- Competing motion signals from on- and off-center retinal ganglion cells (van der Smagt & van de Grind, 1999) reduced the effect (Exp. 1).
- Task-irrelevant motion signals enhanced the effect (Exp. 2).

## References

- Suchow, J. W., & Alvarez, G. A. (2011). Motion silences awareness of visual change. *Current Biology*, 21(2), 140–143. <https://doi.org/10.1016/j.cub.2010.12.019>
- van der Smagt, M. J., & van de Grind, W. A. (1999). Integration and segregation of local motion signals: the role of contrast polarity. *Vision Research*, 39(4), 811–822.
- Wu, Q., & Flombaum, J. I. (2024). The Motion-Silencing Illusion Depends on Object-Centered Representation. *Psychological Science*, 35(5), 504–516. <https://doi.org/10.1177/09567976241235104>